

Statistical Optimization: Lecture 5

Optimization for Exponential Family

Zijian Guo

Zhejiang University
Center for data science

March 19, 2026

Exponential Family

- Many common statistical models belong to the exponential family, including:
 - Binomial
 - Poisson
 - Normal
 - Multinomial
- We begin with the **one-parameter** case and then extend the framework to the **multi-parameter** case.
- Key properties of exponential family: the log-likelihood is convex.

One-Parameter Exponential Family: Definition

Definition The class of distributions $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is said to belong to the **one-parameter exponential family** if there exist real-valued functions $\eta(\theta)$, $B(\theta)$, $T(x)$, and $h(x)$ such that

$$p(x \mid \theta) = h(x) \exp(\eta(\theta) T(x) - B(\theta)). \quad (1)$$

Common support property: If (1) holds and $h(x)$ does not depend on θ , then the support $\{x : p(x \mid \theta) > 0\}$ does not depend on θ . Indeed, $\exp(\eta(\theta)T(x) - B(\theta)) > 0$ always, so positivity of $p(x \mid \theta)$ is determined by $h(x)$.

Example: Bernoulli

If $X \sim \text{Bernoulli}(\theta)$ with $0 < \theta < 1$, then

$$\mathbb{P}(X = x \mid \theta) = \theta^x (1 - \theta)^{1-x} = (1 - \theta) \exp \left(x \log \frac{\theta}{1 - \theta} \right), \quad x = 0, 1. \quad (2)$$

Thus $h(x) = 1$, $T(x) = x$, $\eta(\theta) = \log \frac{\theta}{1 - \theta}$, and

$$B(\theta) = -\log(1 - \theta).$$

Example: Binomial

If $X \sim \text{Binomial}(n, \theta)$ with $0 < \theta < 1$, then

$$\mathbb{P}(X = x \mid \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x} = \binom{n}{x} (1 - \theta)^n \exp \left(x \log \frac{\theta}{1 - \theta} \right), \quad x = 0, 1, \dots, n. \quad (3)$$

Thus $h(x) = \binom{n}{x}$, $T(x) = x$, $\eta(\theta) = \log \frac{\theta}{1 - \theta}$, and

$$B(\theta) = -n \log(1 - \theta).$$

Example: Poisson

If $X \sim \text{Poisson}(\theta)$ with $\theta > 0$, then

$$\mathbb{P}(X = x \mid \theta) = \frac{\theta^x e^{-\theta}}{x!} = \frac{1}{x!} \exp(x \log \theta - \theta), \quad x = 0, 1, 2, \dots \quad (4)$$

Thus $h(x) = 1/x!$, $T(x) = x$, $\eta(\theta) = \log \theta$, and $B(\theta) = \theta$.

Canonical One-Parameter Exponential Family

With the **natural parameter** $\eta = \eta(\theta)$, we define the **canonical one-parameter exponential family**

$$p(x | \eta) = h(x) \exp(\eta T(x) - A(\eta)), \quad (5)$$

where the **log-partition function** $A(\eta)$ is determined by normalization:

$$A(\eta) = \log \left(\int h(x) \exp(\eta T(x)) dx \right) \quad (6)$$

in the continuous case, and the integral is replaced by a sum in the discrete case.

Define the **natural parameter space**

$$\Xi = \{\eta : A(\eta) < \infty\}.$$

Example: Poisson in Canonical Form

For $X \sim \text{Poisson}(\theta)$, let $\eta = \log \theta$. Then

$$p(x \mid \eta) = \frac{1}{x!} \exp(x\eta - \mathbf{e}^\eta), \quad A(\eta) = \mathbf{e}^\eta, \quad \Xi = \mathbb{R}.$$

From (10), $\mathbb{E}_\eta[X] = A'(\eta) = \mathbf{e}^\eta = \theta$, consistent with the mean of a Poisson.

Maximum Likelihood Estimation (MLE) and Convexity

Let $X_1, \dots, X_n$ be i.i.d. from (5). The likelihood is

$$\begin{aligned} L(\eta) &= \prod_{i=1}^n p(x_i | \eta), & \ell(\eta) &= \log L(\eta) = \sum_{i=1}^n \log p(x_i | \eta). \\ -\ell(\eta) &= -\sum_{i=1}^n (\log h(x_i) + \eta T(x_i) - A(\eta)) \\ &= -\sum_{i=1}^n \log h(x_i) - \eta \sum_{i=1}^n T(x_i) + nA(\eta). \end{aligned} \tag{7}$$

Dropping the constant $-\sum_i \log h(x_i)$, the MLE is equivalent to

$$\min_{\eta \in \Xi} \left\{ nA(\eta) - \eta \sum_{i=1}^n T(x_i) \right\}. \tag{8}$$

Thus, if $A(\eta)$ is convex, then (8) is a convex optimization problem.

MGF and Moments $\Rightarrow$ Convexity

Theorem(MGF and moments) Consider the canonical one-parameter exponential family (5). Assume η is an interior point of Ξ . Then for all s in a neighborhood of 0 such that $\eta + s \in \Xi$, the moment generating function (MGF) of $T(X)$ exists and

$$M(s) = \mathbb{E}_{\eta}[e^{s^T T(X)}] = \exp(A(\eta + s) - A(\eta)). \quad (9)$$

Moreover,

$$\mathbb{E}_{\eta}[T(X)] = A'(\eta), \quad \text{Var}_{\eta}(T(X)) = A''(\eta) \geq 0, \quad (10)$$

so $A(\eta)$ is convex on Ξ .

Proof

Using (5),

$$M(s) = \int e^{sT(x)} h(x) \exp(\eta T(x) - A(\eta)) dx = e^{-A(\eta)} \int h(x) \exp((\eta + s)T(x)) dx.$$

By (6), the integral equals $\exp(A(\eta + s))$ (whenever $\eta + s \in \Xi$), hence (9) follows. Differentiating (9) at $s = 0$ gives

$$M'(0) = \mathbb{E}_\eta[T(X)] = A'(\eta), \quad M''(0) = \mathbb{E}_\eta[T(X)^2] = A''(\eta) + (A'(\eta))^2,$$

so $\text{Var}_\eta(T(X)) = M''(0) - M'(0)^2 = A''(\eta) \geq 0$.

Outline

- Multi-parameter Exponential Families

Multi-parameter Exponential Family

Definition The family $\{P_\theta : \theta \in \Theta\}$ is said to be a k -parameter exponential family if there exist real-valued functions $\eta_j(\theta)$, $B(\theta)$, $T_j(x)$, and $h(x)$ ($j = 1, \dots, k$) such that

$$p(x \mid \theta) = h(x) \exp \left(\sum_{j=1}^k \eta_j(\theta) T_j(x) - B(\theta) \right). \quad (11)$$

Example: Normal distribution with unknown mean and variance

Suppose $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$. The joint density is

$$\begin{aligned} p(x \mid \mu, \sigma^2) &= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 + \frac{\mu}{\sigma^2} \sum_{i=1}^n x_i - \frac{n\mu^2}{2\sigma^2} \right\}. \end{aligned} \tag{12}$$

Thus it is a 2-parameter exponential family with sufficient statistics $T_1(x) = \sum_{i=1}^n x_i$ and $T_2(x) = \sum_{i=1}^n x_i^2$.

Example: Multinomial (counts form)

Let $Y = (Y_1, \dots, Y_k) \sim \text{Multinomial}(n; p_1, \dots, p_k)$ with $\sum_{i=1}^k Y_i = n$ and $\sum_{i=1}^k p_i = 1$.
Then

$$\mathbb{P}(Y = y \mid p) = \frac{n!}{\prod_{i=1}^k y_i!} \prod_{i=1}^k p_i^{y_i}.$$

Take $\eta_i = \log(p_i/p_k)$ for $i = 1, \dots, k-1$ and $T_i(y) = y_i$. Then this is a canonical $(k-1)$ -parameter exponential family with base measure $h(y) = \frac{n!}{\prod_{i=1}^k y_i!}$ and

$$A(\eta) = n \log \left(1 + \sum_{i=1}^{k-1} e^{\eta_i} \right).$$

Canonical Multi-parameter Form

Similarly, the canonical form is obtained by setting $\eta = (\eta_1(\theta), \dots, \eta_k(\theta))$:

$$p(x \mid \eta) = h(x) \exp(\langle \eta, T(x) \rangle - A(\eta)), \quad (13)$$

where $T(x) = (T_1(x), \dots, T_k(x))$ and

$$A(\eta) = \log \left(\int h(x) \exp(\langle \eta, T(x) \rangle) dx \right), \quad (14)$$

with sum/integral depending on the model.

Define the **natural parameter space** $\Xi = \{\eta : A(\eta) < \infty\}$.

Basic Properties in the Multi-parameter Case

Theorem For the canonical multi-parameter exponential family (13):

- (a) Ξ is convex;
- (b) $A : \Xi \rightarrow \mathbb{R}$ is convex;
- (c) If η_0 is an interior point of Ξ , then for any s with $\eta_0 + s \in \Xi$,

$$\mathbb{E}_{\eta_0} [e^{\langle s, T(X) \rangle}] = \exp(A(\eta_0 + s) - A(\eta_0)),$$

and (whenever differentiation is justified)

$$\nabla A(\eta) = \mathbb{E}_{\eta}[T(X)], \quad \nabla^2 A(\eta) = \text{Cov}_{\eta}(T(X)) \succeq 0.$$

Lemma: Hölder's Inequality

Lemma(Hölder's Inequality) For $u(x), v(x), h(x) \geq 0$ and $r, s > 1$ with $\frac{1}{r} + \frac{1}{s} = 1$,

$$\int u(x)v(x)h(x) dx \leq \left(\int u(x)^r h(x) dx \right)^{1/r} \left(\int v(x)^s h(x) dx \right)^{1/s}.$$

Remark:(Cauchy-Schwarz). The special case $r = s = 2$ gives the Cauchy-Schwarz inequality.

Proof of (a)–(b) (1/2)

Let $\eta_1, \eta_2 \in \Xi$ and $\alpha \in [0, 1]$. Consider

$$\exp(A(\alpha\eta_1 + (1 - \alpha)\eta_2)) = \int h(x) \exp(\langle \alpha\eta_1 + (1 - \alpha)\eta_2, T(x) \rangle) dx.$$

Rewrite the integrand as

$$h(x) \underbrace{\exp(\langle \eta_1, T(x) \rangle)^\alpha}_{u(x)} \underbrace{\exp(\langle \eta_2, T(x) \rangle)^{1-\alpha}}_{v(x)}.$$

Apply Hölder with $r = 1/\alpha$ and $s = 1/(1 - \alpha)$ (interpret endpoints by continuity):

$$\exp(A(\alpha\eta_1 + (1 - \alpha)\eta_2)) \leq \left(\int h(x) \exp(\langle \eta_1, T(x) \rangle) dx \right)^\alpha \left(\int h(x) \exp(\langle \eta_2, T(x) \rangle) dx \right)^{1-\alpha}.$$

Proof of (a)–(b) (2/2)

Taking logs yields

$$A(\alpha\eta_1 + (1 - \alpha)\eta_2) \leq \alpha A(\eta_1) + (1 - \alpha)A(\eta_2).$$

Thus A is convex on Ξ .

Since the right-hand side is finite, the left-hand side is finite as well. Hence

$$\alpha\eta_1 + (1 - \alpha)\eta_2 \in \Xi,$$

so Ξ is convex.